

GEOOptimize

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Proposal for:	Proposal Date:	Proposal Number:
Number TEN Architectural Group 115 Bannatyne Ave. Suite 310 Winnipeg, MB R3B 0R3 ATTN: Gabe Derksen	2023-11-06	QU-0117
		Reference
		CA MB No 10 WON Arena

Understanding of project:

The community of Wauzhusk Onigum Nation (WON) is looking to build a new net zero certified sportsplex with 700 seat capacity arena, fitness area, and track. The ice plant will feature heat reclaim and the new building is looking to utilize a ground sourced heat pump system, and other technologies to achieve its net zero certification.

Scope of work:

GEOOptimize proposes to work with the design team to develop the design of a horizontal or vertical ground heat exchanger (GHX) for the proposed GSHP system. The work is anticipated to include:

- Development of an hourly energy model of the proposed building, working in collaboration with the SMS or Integrated Design Inc. energy modeler and the design team to optimize the building and systems for operation with a GSHP system.
- Work with SMS on the development of a basis of design for the mechanical system and development of the mechanical system schematic integrated with the GHX for the project.
- Work with SMS on a control strategy and required supplementary heat rejection strategy to mitigate risk of GHX overheating.
- Development of the design of a preliminary design for the GHX for the proposed building.
- Participate in up to three onsite/ in community design review meetings during design development.
- Participate in up to six design meeting with SMS and the design team either virtually or in the City of Winnipeg during Design Development.
- Creation of Contract Documents including IFC drawings and specifications for the GHX design up to and including the GHX manifold inside the building (the point of delineation between the GHX and the mechanical system by SMS).
- Support the bidding process by answering questions and helping to review the qualifications of geothermal sub-contractors.
- Support the construction phase of the project by answering RFI's, attending site meetings, and reviewing site conditions as they pertain to the GHX.

Separate Price: Horizontal Thermal Conductivity Testing

Once complete, the geotechnical report can provide enough information to reasonably estimate the thermal conductivity and diffusivity of either a vertical or horizontally configured GHX. This allows the

calculation of the size and performance of either type of heat exchanger with reasonable accuracy. Conducting a sensitivity analysis based on the anticipated maximum and minimum thermal properties allows us to estimate the expected range of cost that can be expected if a GSHP system is installed.

After the decision is made to proceed with the installation of a GSHP system and the type of GHX to install, the appropriate conductivity test will be used to confirm the assumptions made during the feasibility study. This approach helps reduce the cost of conducting a conductivity test unnecessarily.

A proposal to conduct a vertical thermal conductivity test is already in place from SMS Engineering. If a horizontal GHX is selected as the best option for this project, GEOOptimize can perform a thermal conductivity test for this type of GHX. In this case, GEOOptimize will work with a community excavating contractor or other third-party excavating contractor to excavate two to three test pits in the location of the proposed GHX to collect soil samples and test the thermal conductivity and diffusivity to confirm the GHX design. (NOTE: the cost for the contractor to excavate test pits is NOT included in this proposal).

Design Methodology:

The performance of a GHX is dependent on:

- Total annual heat transfer to and from the building(s)
- Peak instantaneous heat transfer to and from the building(s)
- Ambient earth temperature at the location
- Thermal characteristics of the earth in which the GHX is constructed.
- Detailed design and configuration of the GHX

Thermal properties of the soil can be estimated with reasonable accuracy if information about the geological conditions is available. Information about the geological conditions is often available on water well databases at government operated websites, or in interviews with drilling contractors and / or geologists familiar with the location.

The land area available for construction of a GHX is determined with feedback from the client and development design team. The size and configuration of the land area available for construction of a GHX may determine the length and depth of horizontal boreholes required to meet the heating and cooling loads of the project, or if another type of GHX may be appropriate on the site.

GEOOptimize uses Ground Loop Design (GLD 2016) software to simulate the performance of vertical and horizontal GHX's.

Required information:

The following information is required as it becomes available to accurately model the performance of the GHX:

- Hourly energy model(s) of the building provided in a .csv file that excludes the electrical energy needed to power the heat pump compressors.
- Site plan clearly defining the land area available for construction of the proposed GHX including site services to be installed.
- Building drawings specifications and any available information regarding occupancy and assumed use schedules.
- Information regarding the fill of the nearby 'assumed bog areas.'
- Geotechnical report.

Pricing

Description	Qty	Line Total
Phase 1: Design Development (approx. 4 Months)	1	\$32,400
Phase 2: Construction Documents (approx. 6 Months)	1	\$52,500
Phase 2A: Horizontal TC Test (if required)	1	\$3,000
Phase 3: Bid & Award (approx. 6 Weeks)	1	\$ 5,600
Phase 4: Contract Administration (approx. 15 Months)	1	\$32,200
Total (plus HST)		\$125,700

Terms:

Additional work out of scope will be charged at \$225/hr. Invoices will be submitted monthly for work completed to date. Payment is expected on receipt of invoice.

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